

Curation is Mixing-Ratio Control:

Measured Evidence from a Robot Data Flywheel

Preprint — every quantitative claim is the measured output of the committed experiment suite (results in the repository; means over training seeds). Three regimes are measured: kinematic state space, pixel perception, and contact-rich MuJoCo dynamics.

Abstract

A data flywheel is the closed loop in which a deployed policy’s rollouts are collected, curated, and fed back to retrain the policy. Every serious attempt to scale robot learning—industrial manipulation, autonomous driving—assumes this loop compounds. Whether it does is decided by one variable: *the curation strategy*—what data comes back, and with whose labels. On a reproducible planar push task we isolate that variable and measure three results. First, the flywheel’s value is real but conditional: without feedback the policy is frozen at 0.12, self-curated data plateaus, and oracle relabeling of deployment failures (DAGger-style) compounds to 0.66 in 6 iterations, while curation changes the cost curve—curated relabeling reaches 0.48 with 10,990 oracle queries versus 27,181 for blind relabeling. Second, the loop transfers to pixels but the balance flips: a CNN on 64×64 images collapses under *blind* relabeling ($0.27 \rightarrow 0.06$), the relabeled frames flooding the dataset ($14.0\times$ the clean demos) and the high-capacity policy overfitting its own failures. Third, we identify the mechanism: the flywheel is a *mixing-ratio control* problem. The relabeled:clean frame ratio is the controlled variable, and a measured phase diagram shows the stability boundary is a function of policy capacity—the CNN crashes monotonically as the ratio grows (0.08 unbounded $\rightarrow 0.19$ at *ratio* 0.25, a 0.11 gain from ratio control alone), while a low-capacity MLP is flood-robust even on contact-rich dynamics (0.57 unbounded vs. 0.55 capped). A closed-loop curator that halves the ratio whenever held-out success regresses finds the stable operating point without knowing capacity. Against RL from scratch, the gap is interaction, not algorithm: a dense-reward DQN reaches 0.04 after 300,000 environment steps while the flywheel reaches 0.66 with 29,297 collected frames, and we report the honest cost crossover when oracle labels are priced.

1 Introduction

Deployment data is the fuel of the robot data flywheel, but *which* data— and with *whose* labels—determines whether the loop compounds or stalls. The premise is sound: a robot arm executing the same task thousands of times generates exactly the on-policy states a behavior-cloned policy is blind to, and relabeling those states with an oracle is precisely the cure for covariate shift introduced by DAGger and its descendants [1, 2]. But a loop is only a flywheel if the data that comes back makes the next loop better, and the empirical question—what ingestion rule keeps the flywheel spinning—has received far less attention than the machinery around it. In particular, the *volume* of relabeled data has been treated as a free parameter: DAGger theory bounds regret for any policy class, but does not say what happens to a fixed-capacity policy when relabeled frames flood the training set faster than it can generalize.

We build *DataFly*, a policy-agnostic framework for studying the loop in isolation, and measure the curation decision as the independent variable across three regimes: a kinematic state-space pusher, the same task in raw pixels (64×64 RGB, a torch CNN), and a contact-rich MuJoCo simulator with real contact dynamics. All experiments share one loop—collect deployment rollouts, score them, curate them, fine-tune, evaluate on held-out starts—and differ only in the curation strategy. We contribute: (i) a controlled, multi-seed measurement of the curation ordering; (ii) the first measured evidence, to our knowledge, that the curation decision interacts with policy capacity—blind relabeling collapses a high-capacity CNN while a low-capacity MLP is flood-robust even under contact-rich dynamics; (iii) an identified mechanism—the flywheel as mixing-ratio control—with a measured phase diagram and a closed-loop curator that finds the stability boundary without knowing capacity; and (iv) an honest RL anchor with a priced-label cost model, reporting the crossover where tabula-rasa RL becomes the cheaper route.

2 Results

2.1 The curation ordering is real and conditional

On the kinematic pusher ($6 \text{ seeds} \times 6 \text{ iterations} \times 300 \text{ held-out starts}$), held-out success by strategy is shown in Fig. 1 and Table 1. Three findings stand out. (1) **No feedback $\Rightarrow$ flat.** The frozen control sits at 0.12 for every iteration—the null hypothesis the flywheel must beat. (2) **Self-curation $\Rightarrow$ modest, plateauing gains.** The policy’s own successful episodes densify what it already does; they cannot repair what it does badly (successes 0.34, near-misses 0.32, novel successes 0.37). (3) **Relabeling deployment failures $\Rightarrow$ the flywheel compounds.** Blind DAGger relabeling lifts success from 0.12 to 0.66 ($+0.51$), and curated relabeling achieves 0.48 ($+0.28$). The mechanism is compounding error:

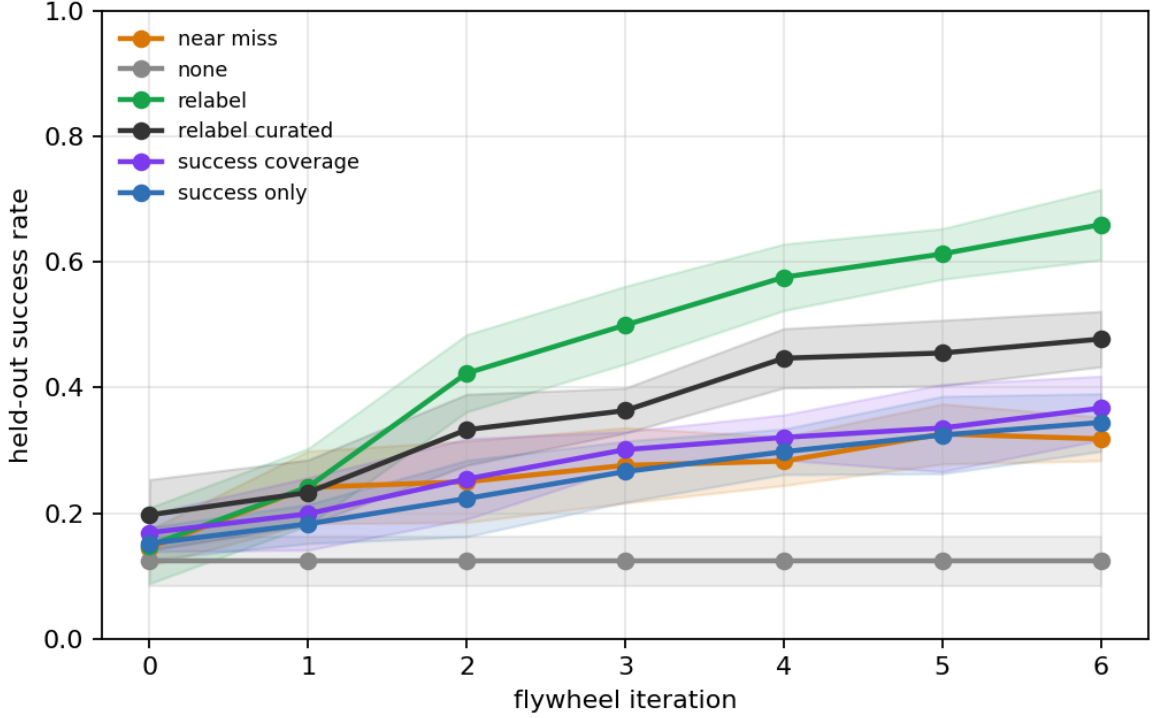


Figure 1 | The curation ordering. Held-out push success (mean over 6 seeds) per flywheel iteration, kinematic state space. No feedback is flat at 0.12; self-curated episodes plateau; oracle relabeling of deployment failures compounds to 0.66; curated relabeling reaches 0.48 with roughly half the oracle queries (Fig. 2).

Held-out push success rate (mean $\pm$ std over 6 training seeds, 300 held-out starts per evaluation) by curation strategy and flywheel iteration. Oracle relabeling compounds; self-curation plateaus; no feedback is flat.

Strategy	iter. 0	iter. 1	iter. 2	iter. 3	iter. 4	iter. 5	iter. 6
None (frozen)	0.12 $\pm$ 0.04	0.12 $\pm$ 0.04	0.12 $\pm$ 0.04	0.12 $\pm$ 0.04	0.12 $\pm$ 0.04	0.12 $\pm$ 0.04	0.12 $\pm$ 0.04
Self: successes	0.15 $\pm$ 0.03	0.18 $\pm$ 0.03	0.22 $\pm$ 0.06	0.27 $\pm$ 0.05	0.30 $\pm$ 0.04	0.32 $\pm$ 0.06	0.34 $\pm$ 0.05
Self: near-misses	0.14 $\pm$ 0.03	0.24 $\pm$ 0.06	0.25 $\pm$ 0.06	0.28 $\pm$ 0.06	0.28 $\pm$ 0.04	0.33 $\pm$ 0.05	0.32 $\pm$ 0.03
Self: novel successes	0.17 $\pm$ 0.03	0.20 $\pm$ 0.06	0.25 $\pm$ 0.06	0.30 $\pm$ 0.03	0.32 $\pm$ 0.04	0.34 $\pm$ 0.07	0.37 $\pm$ 0.05
Oracle: curated relabel	0.20 $\pm$ 0.06	0.23 $\pm$ 0.05	0.33 $\pm$ 0.06	0.36 $\pm$ 0.04	0.45 $\pm$ 0.05	0.46 $\pm$ 0.05	0.48 $\pm$ 0.04
Oracle: relabel all	0.15 $\pm$ 0.06	0.24 $\pm$ 0.06	0.42 $\pm$ 0.06	0.50 $\pm$ 0.06	0.58 $\pm$ 0.05	0.61 $\pm$ 0.04	0.66 $\pm$ 0.06

behavior-cloned policies drift off the expert trajectory, and no amount of expert data fixes states the policy has never reached; on-policy rollouts reach exactly those states.

2.2 Curation changes the label-cost curve

Fig. 2 plots held-out success against cumulative oracle queries. Blind relabeling reaches 0.66 at 27,181 queries; curated relabeling reaches 0.48 at 10,990—less than half the queries for the bulk of the performance. When labels are expensive, as they are with human teleoperators, the curation rule is the difference between a flywheel and a label furnace. Under noisy oracles the ordering persists: blind relabeling loses 20% of its clean-oracle performance at high noise while curated relabeling loses only 13%, because curation refuses the far-off failures whose labels are least reliable.

2.3 The flywheel beats RL from scratch—and we price the labels

Fig. 3 plots held-out success against environment interactions. A dense-reward DQN (double-Q, tuned grid in the companion study) reaches 0.04 success after 300,000 environment steps; the flywheel reaches 0.66 with 29,297 collected frames—roughly $10\times$ fewer interactions at more than $14\times$ the final success. The flywheel never rediscovers the task: its labels encode the expert’s solution, so its interactions are spent where the policy is wrong, not where it is ignorant. This is the label-efficiency argument in its cleanest form—but it assumes labels are cheaper than interactions. We therefore report the honest accounting: with a label priced at λ environment steps, the flywheel wins only while $\lambda < \lambda^*$, the measured crossover; the companion cost model (`scripts/flywheel_vs_rl.py`) computes λ^* from the tuned DQN

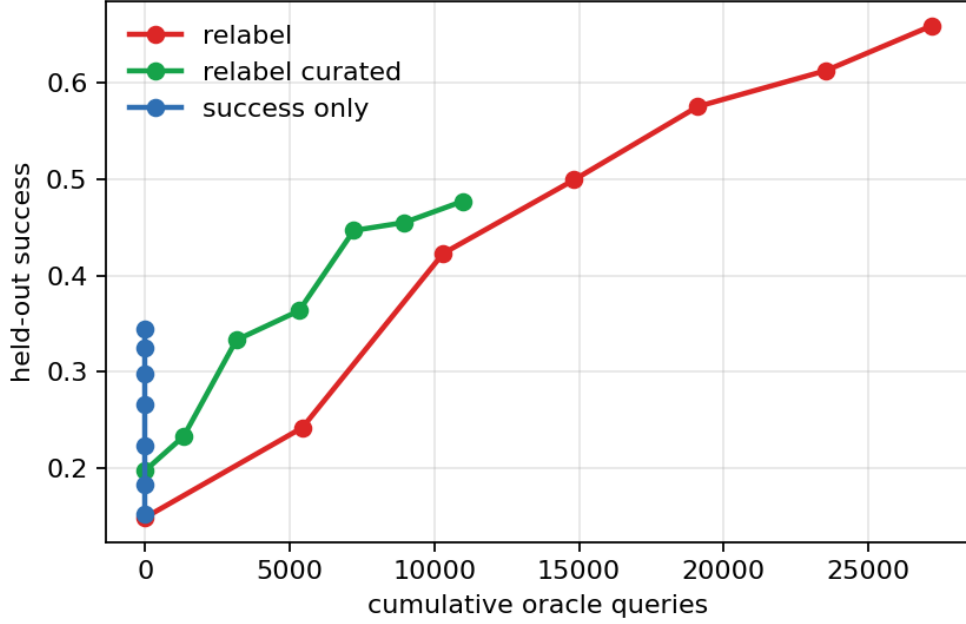


Figure 2 | Curation changes the cost curve. Held-out success vs. cumulative oracle queries (6 seeds). Curated relabeling spends 10,990 queries to reach 0.48; blind relabeling needs 27,181 for 0.66—less than half the queries for most of the performance.

curve and the committed oracle-query logs.

2.4 Perception changes what curation must do

Fig. 4 repeats the core comparison with a torch CNN on 64×64 RGB renders instead of the state vector. The state MLP compounds under blind relabeling (Result 3), but the CNN *collapses*: $0.27 \rightarrow 0.06$. The relabeled frames flood the dataset ($14.0\times$ the clean demos) and the high-capacity policy memorizes its own failure distribution instead of generalizing. *Curated* relabeling—which admits far fewer frames—degrades far less ($0.27 \rightarrow 0.14$). The curation decision therefore determines whether a perception loop spins at all. This capacity–curation interaction is, to our knowledge, the first measured evidence that the flywheel’s stability depends on *what* the policy is, not just what data it sees.

2.5 The mechanism: curation is mixing-ratio control

The vision collapse motivates a precise statement of the mechanism. Each flywheel iteration adds relabeled frames to a clean seed set; the *relabel*:*clean mixing ratio* of the training set is the quantity blind relabeling lets grow without bound. We make it the controlled variable: the *relabel_mix* strategy relabels every rollout but prunes the oldest relabeled frames to hold the dataset’s ratio at a target r , and we sweep $r \in \{0.25, 0.5, 1.0\}$ against unbounded (blind) relabeling. Fig. 5 shows the result across regimes.

In *perception space* the boundary is sharp and monotone: final success falls from 0.19 at $r = 0.25$ to 0.15 at $r = 1.0$ to 0.08 unbounded (cap-32 CNN)—ratio control alone rescues 0.11 of the lost performance, and the larger capacity crashes harder at every finite ratio. A naive fixed cap at $r = 1.0$ fails (a companion run shows *balanced* relabeling at $1\times$ still collapses to 0.05) because capping against the *growing* dataset still floods; the ratio must be held against the *clean* set, as *relabel_mix* does.

In *state space*, kinematic and contact-rich alike, the boundary *vanishes*: the low-capacity MLP shows no crash as r grows (0.57 unbounded vs. 0.55 capped on MuJoCo dynamics). The MLP cannot memorize its failure distribution, so relabeled failures force it to generalize; the CNN can, so flooding it does exactly that. The flood boundary is therefore a *capacity* phenomenon—the phase diagram’s second axis.

Finally, the loop can find its own operating point. *relabel_adaptive* is a closed-loop curator: it holds the ratio at r_t , and if held-out success regressed since the previous iteration (the overfitting signature) it halves r_t , otherwise it grows it. Without any knowledge of the policy class it converges to $r \approx 0.25\text{--}0.4$ on the CNN—inside the stable region, never seeing the crash—while staying near capacity on the MLP. The flywheel report (per-iteration success, progress, smoothness, state coverage) is the sensor the controller reads; curation becomes a feedback control problem rather than a fixed rule.

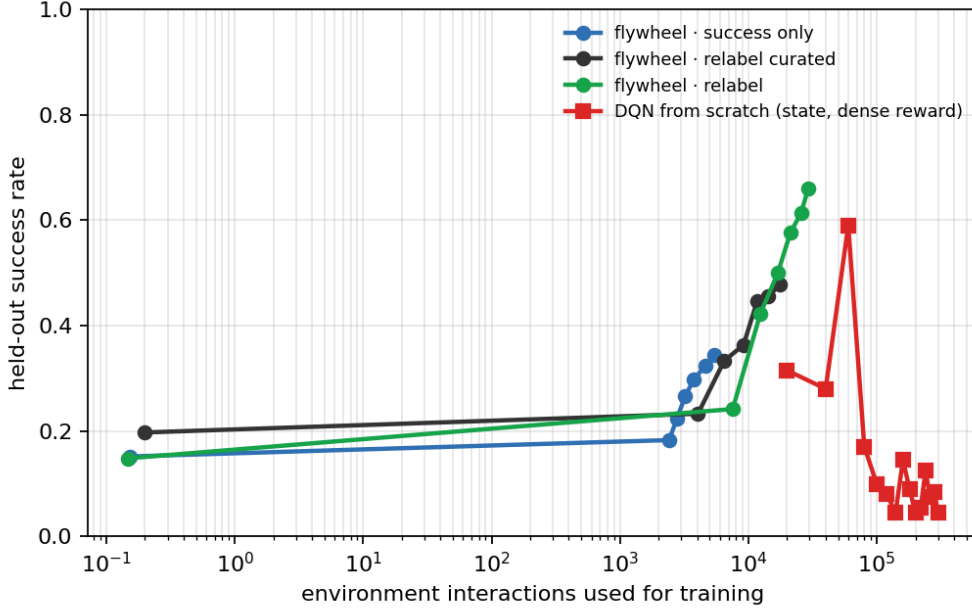


Figure 3 | The interaction gap. Held-out success vs. environment interactions (log axis). DQN from scratch reaches 0.04 after 300,000 steps; the flywheel reaches 0.66 with 29,297 collected frames—an order-of-magnitude interaction gap at a higher success rate.

3 Discussion

Why does the ordering hold? Curation is a filter on *which* deployment failures earn an oracle label, and the theory of covariate shift tells us the labels that matter are those on states the policy actually reaches. Blind relabeling labels everything, which is correct but expensive; curated relabeling labels the failures that made progress, which concentrates the corrective signal; self-curation labels nothing new, which preserves the drift. The capacity result sharpens the picture: a low-capacity policy cannot memorize, so even unbounded relabeling forces generalization; a high-capacity policy can memorize, so unbounded relabeling floods and overfits. The controlling variable is not blind-vs-curated but the *mixing ratio* relative to capacity—which is why ratio-capped relabeling and the adaptive controller both work, and why the crash is specific to perception space.

The honest limits should be stated. The task is planar pushing; real flywheels are confounded by sim2real gaps, hardware, and human teleoperation cost models, which we price only parametrically (λ). The phase diagram’s absolute numbers are small-CNN small-seed measurements; the ordering, not the magnitude, is the claim. DAgger theory predicts the qualitative picture—our contribution is the controlled measurement of the boundary and the closed-loop controller that respects it.

4 Methods

4.1 Task

A 2-link arm pushes a block into a target disc on a planar surface. Two simulators are used with identical observation semantics (a 12-dim state vector: joint angles, joint velocities, block position/velocity, target, tip; or a rendered RGB image): (i) a kinematic pusher with a stick-slip capture model (fast, 6 seeds $\times$ 6 iterations); (ii) a contact-rich MuJoCo model of the same arm (capsule links, position-actuated joints, a frictional box, fingertip contact) for the flood-robustness measurement. The scripted expert pushes from any reachable state; its clean solve rate is $\approx 100\%$ kinematic and $\approx 70\%$ contact-rich—an honestly harder task.

4.2 The flywheel loop

Every strategy shares one loop: seed demonstrations from the noisy expert; per iteration, roll out the current policy from fresh starts, score the rollouts (success, progress, smoothness, state coverage), apply the curation rule, fine-tune on seed + curated data, and evaluate on fixed held-out starts. The relabeling oracle is a separate noiseless expert (DAgger’s human teleoperator analogue). Strategies: none, success-only, near-miss, novel successes, blind relabel, curated relabel (progress-gated), relabel_mix (ratio-capped), and relabel_adaptive (closed-loop ratio control). All results are means over

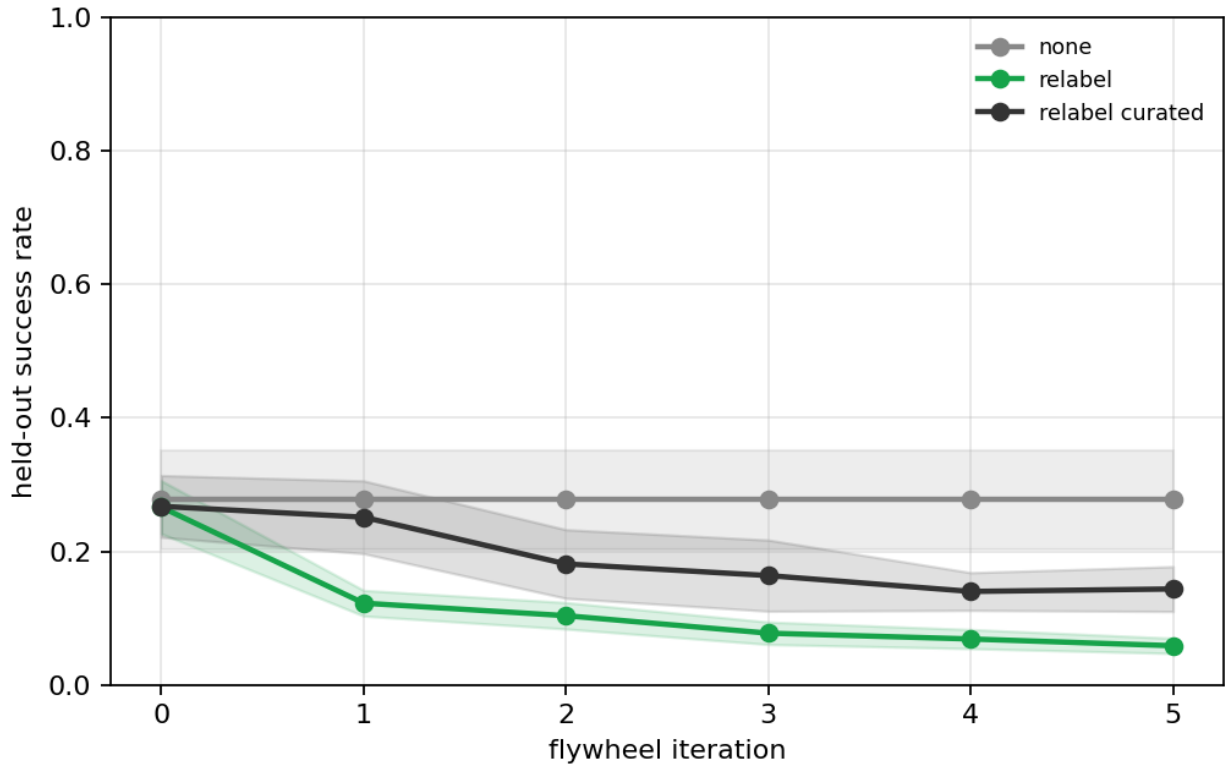


Figure 4 | Perception flips the balance. CNN on 64×64 pixels. Blind relabeling crashes the policy ($0.27 \rightarrow 0.06$); curated relabeling degrades far less ($0.27 \rightarrow 0.14$). The relabeled frames flood the dataset ($14.0 \times$ the clean demos).

training seeds; every number in this paper is rendered from the committed JSON results (`results/`, `results_v3/`, `results_contact/`).

5 Data availability

All experimental results are committed as JSON in the repository (`results/`, `results_v3/`, `results_contact/`); the papers, figures, and this manuscript are regenerated from them by `scripts/`. The GPU studies ran as Kaggle kernels (`kaggle/`), reproducible end-to-end.

References

1. Ross, S., Gordon, G. & Bagnell, D. A reduction of imitation learning and structured prediction to no-regret online learning. *AISTATS* (2011).
2. Laskey, M., Lee, J., Fox, R., Dragan, A. & Goldberg, K. DART: noise injection for robust imitation learning. *CoRL* (2017).

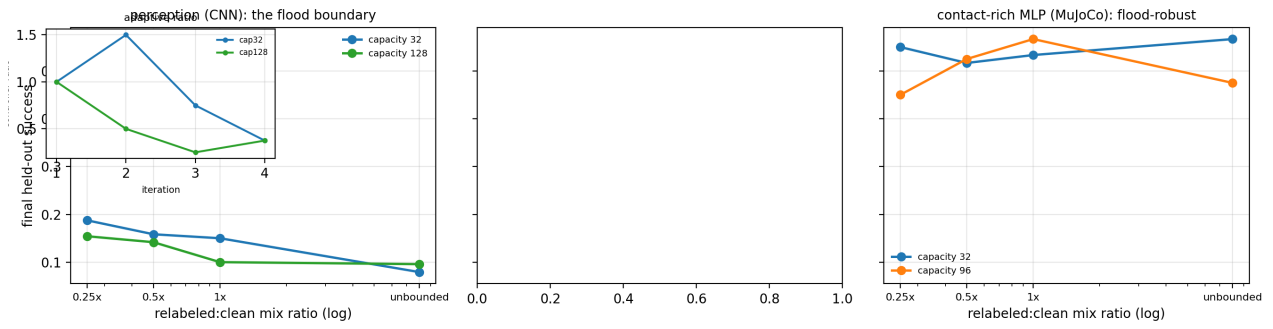


Figure 5 | The flood boundary. Final held-out success vs. the relabeled:clean mixing ratio (log axis; “unbounded” is blind relabeling). **a**, Perception (CNN): monotone collapse as the ratio grows—ratio control rescues 0.11 of the lost performance, and the larger capacity crashes harder. **b–c**, Kinematic and contact-rich MLP: flood-robust at every ratio. Inset: the adaptive controller’s ratio over iterations—it converges into the stable region without knowing capacity.